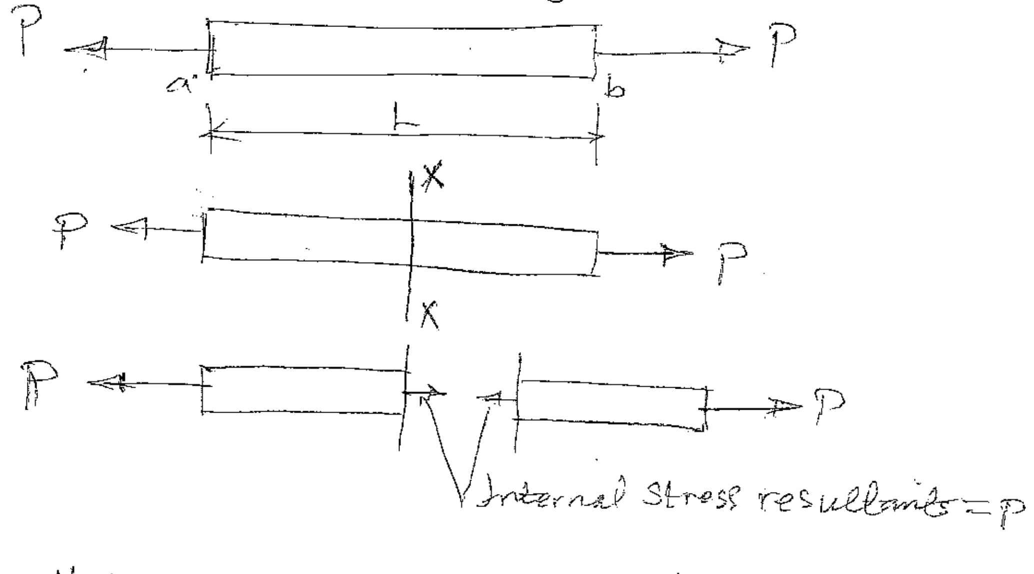
STRESS AND STRAIN :

STRESS:

The internal resistance offered by a body or material due to externally applied force or load is known as stress; OR, The internal resistance developed by a body over a unit area of its cross-section.

Whenever a body is subjected to forces, the body deforms. The bodies/material in practice are not rigid. The body resists the deformation by developing stresses. The body reaches an equilibrium position when the internal stress resultant equals the external force. The deformation of the body and the stress can be related for any body depending upon the its material properties.

Consider the effect of two axial forces P acting on a rod, which tend to stretch or elongate the rod.



Mathematically, $\text{Stress}(\sigma) = \frac{\text{Load}}{\text{Area}} = \frac{P}{A}$ ——— 1.1

The symbol Sigma (σ) is used to denote normal stress.

Types of stresses:

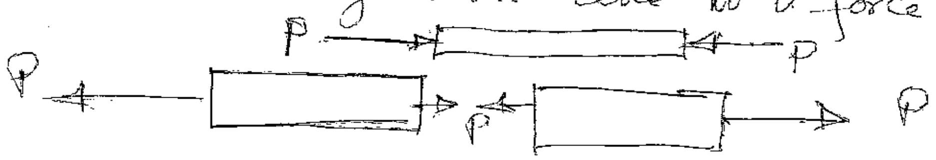
There are basically two types of stresses, normal and tangential (shear) stress.

Normal or Direct Stress.

It is a stress acting perpendicular to the cross-section of the member. In this the direct force is uniformly or equally applied across the cross-section, then the internal forces set up are also distributed uniformly and the ~~bar~~ is said to be a uniform direct or normal stress. Normal stress can be of two types - tensile stress and compressive stress

Tensile stress:

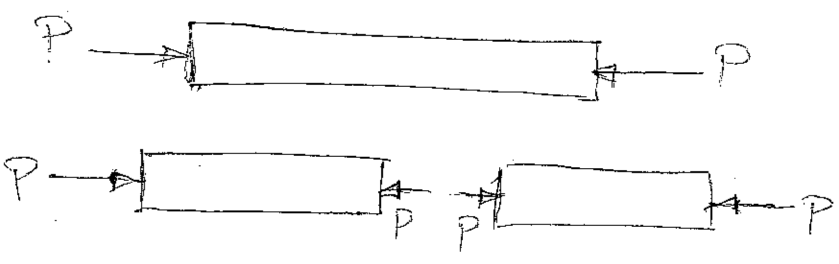
It is the normal stress acting in such a direction that it resists the elongation of a member i.e the effect tends to stretch or elongated.. due to a force of pull.



Tensile Stress

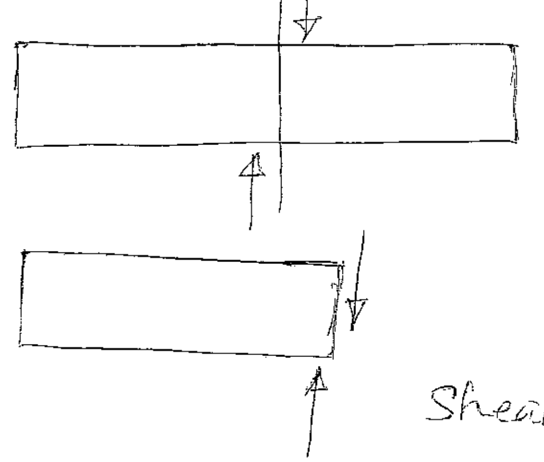
Compressive stress:

It is the normal stress acting at a section of the body away from each other. The external forces tend to shorten or compress the body



Tangential Stress :

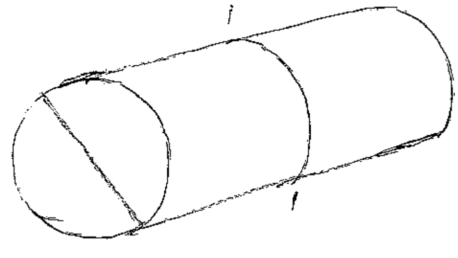
If the external forces tend to shear the member along the cross-section, internal forces (stress) developep act tangentially to the cross-section. Tangential or Shear stress is denoted by the Symbol τ (tau).



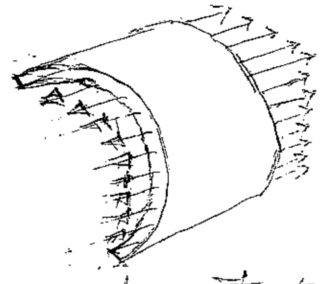
Shearing Stress

Longitudinal stress:

If we consider a section of the cylinder, the internal pressure acting on the ends causes a stretching of the fibres. The forces acting on the ends requires forces to be developed by stresses acting normal to the section as shown. These stresses are called longitudinal stresses as they act along the length of the cylinder. Note that they act normal to the section and the same as normal stresses.



Cylinder under internal pressure



longitudinal stress

Hoop Stress:

These are also normal stresses but they act along the circumference of the cylinder, they are called hoop or circumferential stress.

hoop stress

Unit of Stress:

The basic unit of stress is N/m^2. This unit is also known as Pascal (Pa). Depending on the magnitude of the stress we can use units like N/mm^2, N/m^2, KN/m^2, KN/mm^2 etc. Multiples of the basic unit (N/m^2 or Pa) can be used like megapascal (MPa), gigapascal (GPa) etc.

$1\,MPa = 10^6\,Pa$ & $1\,GPa = 10^9\,Pa$

Also note that MPa & N/mm^2 are numerically equal

$$1\,MPa = 10^6\,N/m^2 = \frac{10^6\,N}{10^6\,mm^2} = 1\,N/mm^2$$

STRAIN

Is defined as a measure of the deformation of the body produced by the load.

If a bar material is subjected to a direct load and hence a stress, the bar material will change in length. If the bar has an original length L and changes in length by an amount δL, the strain produced is given as follows:

$$\text{Strain}(\varepsilon) = \frac{\text{Change in length}}{\text{original length}} = \frac{\delta L}{L} \quad (1.2)$$

The Symbol of Strain is ε (epsilon)

Strain is a non-dimensional quantity ie it has no units, it is simply a ratio of two quantities with the same unit.

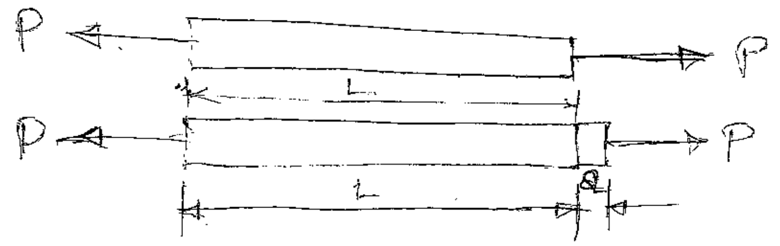
Just as there are two types of stresses, there are also two types of strains) They are normal strain and tangential strain.

Normal strain:

Normal strain can be tensile or compressive.

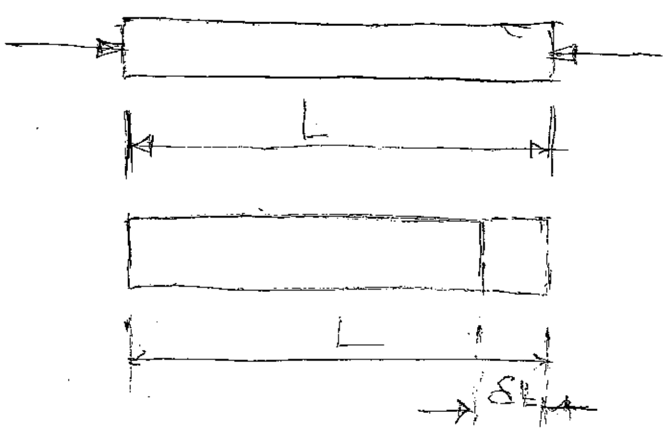
Tensile strain:

In this the body undergoes elongation under tensile load:



Compressive strain.

In this case the body reduces in length due to external forces.

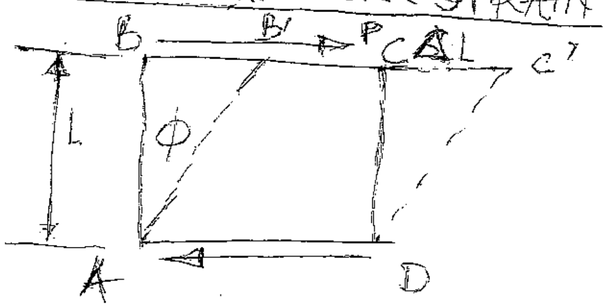


In practice, the extensions of materials under load are very small, it is often convinient to measure the strains in the form of strain $\times 10^{-6}$, ie microstrain and the symbol used becomes $\mu\varepsilon$.
Alternatively, strain can be expressed as a percentage strain

$$\text{ie strain } (\varepsilon) = \frac{\delta L}{L} \times 100\% \quad \text{—— (1.3)}$$

Tensile stresses and strains are considered positive in a sense producing increase in length while compressive stresses and strains are considered negative in a sense producing a decrease in length.

TANGENTIAL OR SHEAR STRAIN



The situation above shows a force P is applied tangential to the top face of the block. The force tends to strain the bar as shown. From the triangle Δ ABB', the shear strain is the angle ϕ, ie the distortion produce by the shear force, it can be defined as the change in right angle. From the diagram, we find that

$\tan\phi = \frac{\Delta L}{L}$, as the deformation is generally small

$\tan\phi = \phi$, thus the shear strain $\phi = \frac{\Delta L}{L}$ —— (1.3)

Note that ϕ is non-dimensional.

HOOKE'S LAW:

An important relationship between stress and strain is given by Hooke's Law, which states that: "Stress is proportional to strain so long as the material behaves elastically."

Elasticity is the property of a material to regain its original shape on the removal of an externally applied force. Some materials show elastic properties till they have attained considerable stress, while other materials exhibit such a property only for very small value of stress. The stress up to which the material behaves elastically is known as its elastic limit. Robert Hooke first stated this property of a material, after noticing the linear relationship between applied forces and deformations. Thus stress is proportional to strain within elastic limit.

$$\text{Stress}(\sigma) \propto \text{Strain}(\varepsilon)$$

ie $\dfrac{\text{Stress}}{\text{Strain}} = \text{Constant}$ | Stress = a constant × strain

Modulus of Elasticity (Young's Modulus).

Within the limits for which ~~Hooke's~~ Law is obeyed, th ratio of the direct stress to the strain produced is called Young's Modulus or the Modulus of Elasticity (E)

Thus $$E = \frac{\sigma}{\varepsilon} \quad \text{------} \quad (1.5)$$

$$= \frac{P}{A} \div \frac{\delta L}{L}$$

$$= \frac{PL}{A\delta L} \quad \text{------} \quad (1.6)$$

E is therefore a constant material and is usually assumed to be the same in tension or compression.

Modulus of Rigidity or Shear Modulus (G).

For elastic materials it is found that shear strain is proportional to the shear stress producing it, within certain limits.

$$\tau \propto \phi$$

$$\tau = \text{a constant} \times \phi$$

The constant is known as the Shear Modulus or Modulus of Rigidity and it is usually denoted by the symbol G.

$$\therefore \text{Modulus of Rigidity } (G) = \frac{\text{Shear Stress}}{\text{Shear Strain}}$$

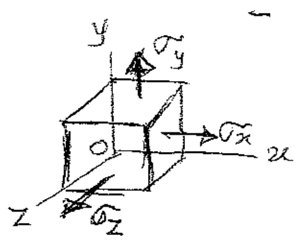
ie $$G = \frac{\tau}{\phi} \qquad \text{———— } (1.7)$$

Bulk Modulus (K).

Bulk modulus is defined in terms of stress and volumetric strain. When a body is subjected to stresses that equal in all direction, the bulk modulus is given by

$$K = \frac{\text{Stress}}{\text{Volumetric strain}}$$

$$= \frac{\sigma}{\varepsilon_v} \qquad \text{———— } (1.8)$$



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HOD

Volumetric Strain.

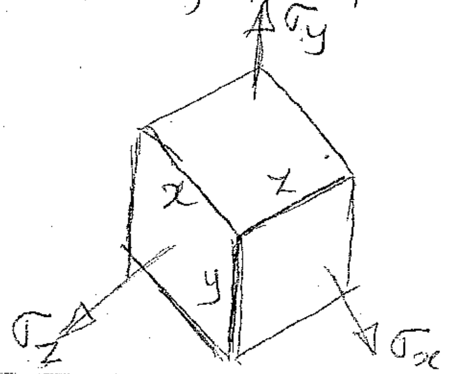
The Volumetric Strain is defined as a change in Volume per unit volume; This is also known as dilatation: the act of being changed or stretched beyond normal

Thus, Volumetric strain, $\varepsilon_v = \frac{\delta V}{V}$ ——— (1.9)

where δV is the change or increase in volume and V is the original Volume

It can be shown that in case of a multi-axial stress situation, Volumetric Strain is equal to the sum of the strains along the three mutually perpendicular directions.

Consider a rectangular solid of sides x, y and z, under the action of Principal stresses σ_x σ_y and σ_z respectively.



Then if ε_x, ε_y and ε_z are the corresponding linear strains, the dimensions becomes $x+\varepsilon_x x$, $y+\varepsilon_y y$ and $z+\varepsilon_z z$

Then, Volumetric Strain $= \frac{\text{Change in Volume}}{\text{original Volume}}$

$$= \frac{x(1+\varepsilon_x)\,y(1+\varepsilon_y)\,z(1+\varepsilon_z) - xyz}{xyz}$$

$$= (1+\varepsilon_x)(1+\varepsilon_y)(1+\varepsilon_z) - 1$$

$$= 1+\varepsilon_x+\varepsilon_y+\varepsilon_z+\varepsilon_x\varepsilon_y+\varepsilon_x\varepsilon_z+\varepsilon_y\varepsilon_z+\varepsilon_x\varepsilon_y\varepsilon_z-1$$

Volumetric strain = $\varepsilon_x + \varepsilon_y + \varepsilon_z$; To sufficient accuracy since the strains are small the volumetric strain reduces to

$$\varepsilon_v = \varepsilon_x + \varepsilon_y + \varepsilon_z \quad (1.10)$$

| $\varepsilon_x\varepsilon_y, \varepsilon_x\varepsilon_z, \varepsilon_y\varepsilon_z$ & $\varepsilon_x\varepsilon_y\varepsilon_z$ are negligible

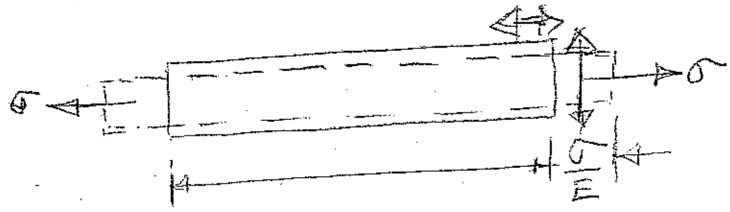
Expressing this in words, the volumetric strain is the algebraic sum of the three principal strains.

Substituting for the strains in terms of the principal stresses

$$\text{Volumetric strain } \varepsilon_v = \frac{(\sigma_x + \sigma_y + \sigma_z)(1-2\nu)}{E} \quad (1.11)$$

POISSON'S RATIO.

If a bar is subjected to a longitudinal stress there will be a strain in this direction equal to $\frac{\sigma}{E}$, there will also be a strain in all directions at right angles to σ, the final shape being dotted.



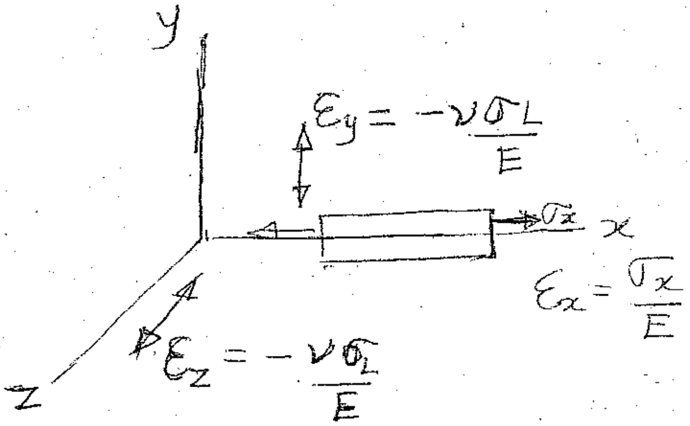
$$\varepsilon = \frac{\sigma}{E}$$

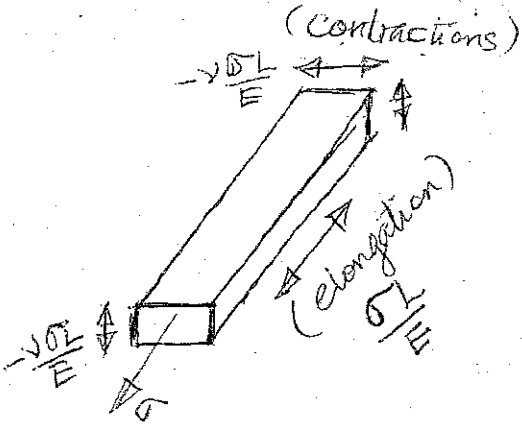
$$\text{or } \frac{\delta L}{L} = \frac{\sigma}{E} \qquad \text{or } \delta L = \frac{\sigma L}{E}$$

It is found that for an elastic material the lateral strain (strains in all directions at right angles) is proportional to the longitudinal strain and is of the opposite type. The ratio of $\frac{\text{lateral strain}}{\text{longitudinal strain}}$ produced is called the Poisson's Ratio and the symbol used is ν. | μ or $\frac{1}{m}$

$$\text{Thus longitudinal strain} = \frac{\sigma}{E} \quad (1.12)$$

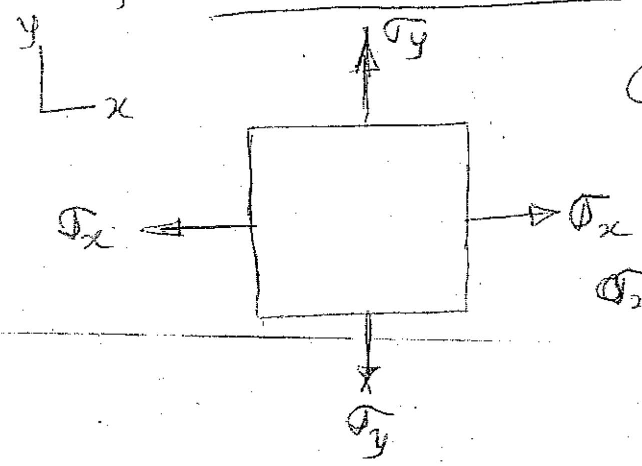
$$\text{but the lateral strain} = -\nu \cdot \frac{\sigma}{E} \quad (1.13)$$





In the figure above σ_x is the stress acting along the x-direction and the strain ε_x along this direction is $\frac{\sigma_x}{E}$, the strains in the lateral directions y and z are $\varepsilon_y = -\nu\frac{\sigma_x}{E}$ & $\varepsilon_z = -\nu\frac{\sigma_x}{E}$

Thus for two-dimensional (biaxial) stress system



Consider the strains produced by each stress separately.

σ_x will cause strain $\frac{\sigma_x}{E}$ in the direction of σ_x and strain $-\nu\frac{\sigma_x}{E}$ in the direction of σ_y.

Similarly, σ_y will cause strain $\frac{\sigma_y}{E}$ in the direction of σ_y and strain $-\nu\frac{\sigma_y}{E}$ in the direction of σ_x. Since the strains are small, the resultant strains are given by the algebraic sum of those due to each stress separately, ie strain in the direction of x;

$$\varepsilon_x = \frac{\sigma_x}{E} - \nu\frac{\sigma_y}{E} \qquad (1.14)$$

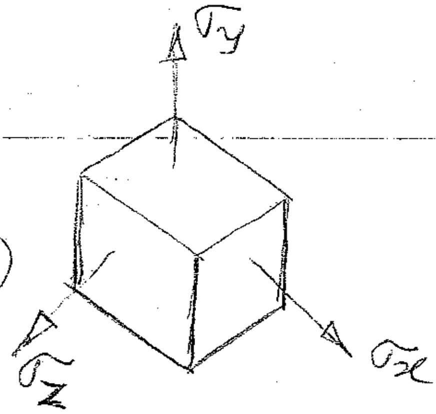
and Strain in the direction of y,

$$\varepsilon_y = \frac{\sigma_y}{E} - \frac{\nu\sigma_x}{E} \qquad (1.15)$$

<u>NOTE</u>: Tensile stress is taken to be positive, and while compressive stress is taken to be negative.

<u>Principal strains in three (multi-axil) Dimensions.</u>

By a similar derivation to the previous case; it can be shown that the principal strains in the directions of σ_x, σ_y and σ_z are:



$$\varepsilon_x = \frac{\sigma_x}{E} - \frac{\nu\sigma_y}{E} - \frac{\nu\sigma_z}{E} \qquad (1.16)$$

$$\varepsilon_y = \frac{\sigma_y}{E} - \frac{\nu\sigma_z}{E} - \frac{\nu\sigma_x}{E} \qquad (1.17)$$

$$\varepsilon_z = \frac{\sigma_z}{E} - \frac{\nu\sigma_x}{E} - \frac{\nu\sigma_y}{E} \qquad (1.18)$$

It should be particularly noted that stress and strain in any given direction are not proportional where stress exists in more than one dimension. In fact strain can exists without stress in the same direction.

eg if $\sigma_z = 0$, then $\varepsilon_z = -\frac{\nu\sigma_x}{E} - \frac{\nu\sigma_y}{E}$ and vice-versa.

<u>RELATIONSHIP BETWEEN THE THREE ELASTIC CONSTANTS: E, G & K.</u>

We need the three elastic constants, E, G and K of a material and its Poisson's ratio ν, to be able to completely calculate the deformations.

Recall that the volumetric strain is given by

$$\varepsilon_v = \frac{(\sigma_x + \sigma_y + \sigma_z)(1-2\nu)}{E}$$

If $\sigma_x = \sigma_y = \sigma_z = \sigma$

then, $$\varepsilon_v = \frac{3\sigma(1-2\nu)}{E}$$

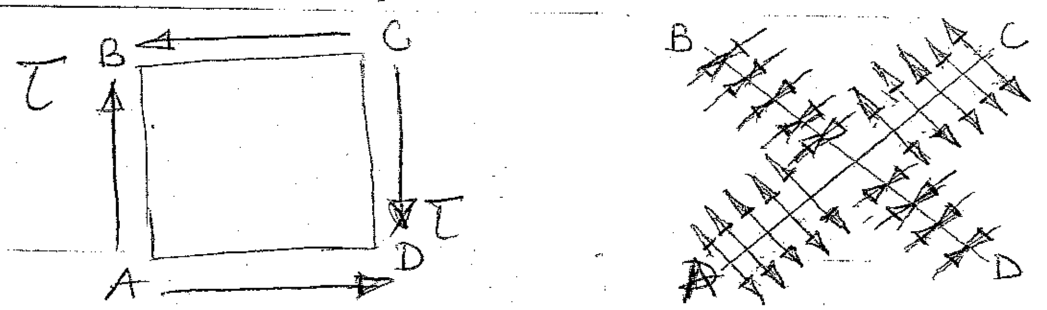
Recall that $K = \frac{\sigma}{\varepsilon_v}$ (by defination)

Therefore $$E = \frac{3\sigma}{\varepsilon_v}(1-2\nu) = 3K(1-2\nu) \quad \text{——} (1.19)$$

OR $$\boxed{K = \frac{E}{3(1-2\nu)}} \quad \text{——} (1.20)$$

This gives the relationship between Young's modulus (E) and bulk Modulus (K)

To derive the relationship between E and G, Consider the case of pure shear stress shown below



It is considered that there are compressive and tensile stresses along the diagonals AC and BD, and these stresses are equal to the shear stress τ. It is also considered that the diagonal strains are each equal to half the shear strain

ie $$\varepsilon_{AC} = \frac{\phi}{2}$$

—/14

Therefore, $\varepsilon_{AC} = \dfrac{\phi}{2} = \dfrac{\tau}{2G}$

Since there is tensile stress along the diagonal AC and compressive stress along the diagonal BD each equal to τ in magnitude, then from the 2nd figure above;

$$\varepsilon_{AC} = \frac{\sigma}{E} + \frac{\sigma\nu}{E} = \frac{\sigma}{E}(1+\nu) = \frac{\tau}{E}(1+\nu)$$

Therefore,

$$\frac{\tau}{2G} = \frac{\tau}{E}(1+\nu)$$

$$\Rightarrow E = 2G(1+\nu) \qquad (1.21)$$

$$\Rightarrow \boxed{G = \frac{E}{2(1+\nu)}} \qquad (1.22)$$

This is the relationship between E and G

Also since $E = 3K(1-2\nu)$

$$\boxed{G = \frac{3K(1-2\nu)}{2(1+\nu)}} \qquad (1.23)$$

This gives the relationship between G & K.

Eliminating ν from these equations, one can get

$$E = \frac{9KG}{(3K+G)} \qquad (1.24)$$

Principal Stresses Determined from Principal Strains:

(a) Three-dimensional stress system

Recall from equations (1.16), (1.17) and (1.18)

$$E\varepsilon_x = \sigma_x - \nu\sigma_y - \nu\sigma_z \quad \text{------} \quad (1.25)$$

$$E\varepsilon_y = \sigma_y - \nu\sigma_z - \nu\sigma_x \quad \text{------} \quad (1.26)$$

$$E\varepsilon_z = \sigma_z - \nu\sigma_x - \nu\sigma_y \quad \text{------} \quad (1.27)$$

and Subtracting (1.26) from (1.25) gives

$$E(\varepsilon_x - \varepsilon_y) = (\sigma_x - \sigma_y)(1 + \nu) \quad \text{------} \quad (1.28)$$

From (1.25) and (1.27), eliminating σ_3 will give

$$E(\varepsilon_x + \nu\varepsilon_z) = \sigma_x(1-\nu^2) - \sigma_y(1+\nu)\nu \quad \text{------} \quad (1.29)$$

Multiplying (1.28) by ν and Subtracting from (1.29), we have

$$E[(1-\nu)\varepsilon_x + \nu(\varepsilon_y + \varepsilon_z)] = \sigma_x(1-\nu-2\nu^2) = \sigma_x(1+\nu)(1-2\nu)$$

Rearranging,

$$\sigma_x = \frac{E[(1-\nu)\varepsilon_x + \nu(\varepsilon_y + \varepsilon_z)]}{(1+\nu)(1-2\nu)} \quad \text{------} \quad (1.30)$$

Similarly

$$\sigma_y = \frac{E[(1-\nu)\varepsilon_y + \nu(\varepsilon_z + \varepsilon_x)]}{(1+\nu)(1-2\nu)} \quad \text{------} \quad (1.31)$$

and

$$\sigma_z = \frac{E[(1-\nu)\varepsilon_z + \nu(\varepsilon_x + \varepsilon_y)]}{(1+\nu)(1-2\nu)} \quad \text{------} \quad (1.32)$$

—/14b

(b) Two-dimensional stress system

$$\sigma_3 = 0$$

$$E\varepsilon_x = \sigma_x - \nu\sigma_y$$

$$E\varepsilon_y = \sigma_y - \nu\sigma_x$$

Solving these equations for σ_x and σ_y gives

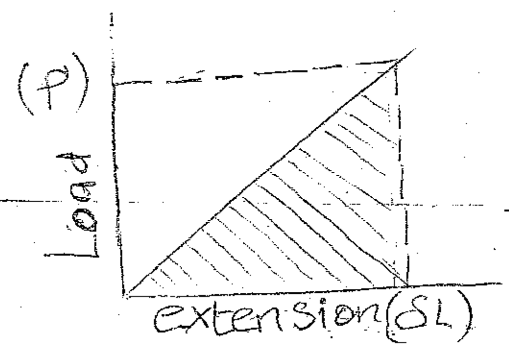
$$\sigma_x = \frac{E(\varepsilon_x + \nu\varepsilon_y)}{1-\nu^2} \quad \text{——— (1.33)}$$

and $$\sigma_y = \frac{E(\nu\varepsilon_x + \varepsilon_y)}{1-\nu^2} \quad \text{——— (1.34)}$$

STRAIN ENERGY.

We are already familiar with the three types of straining actions, namely tensile, compressive and shear strain; they are used to calculate stresses and deformations in structures. We have another basic concept - that of the strain energy - its application are particularly useful and effective in the calculation of deformation in structures subjected to impact loads.

A structure deforms when subjected to loads and the points of application of the load consequently move. The applied loads thus does work, and this work is stored as strain energy in the structure, its function is to help the structure regain or recover its original shape so long as the stresses are within elastic limits.



When a tensile or compressive load P is applied to a bar there is a change in length δL, which for an elastic material is proportional to the load as shown in the figure.

The strain energy (U) is defined as the workdone by the load in straining it.
For a gradually applied or static load the workdone is represented by the shaded area.

Thus, $U = \frac{1}{2} P \delta L$ ——————— (1.35)

This can be expressed in terms of stress and dimensions for a bar of uniform cross-section A and length l we know that $P = \sigma A$ and $\delta L = \varepsilon L = \frac{\sigma L}{E}$

Substituting in equation (1·25), we have

$$U = \frac{1}{2} \cdot \sigma A \cdot \frac{\sigma L}{E} = \frac{1}{2} \cdot \sigma^2 \frac{AL}{E}$$

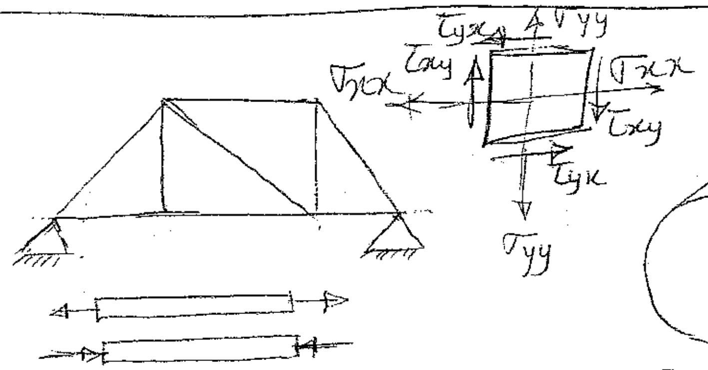
$$= \frac{\sigma^2 AL}{2E} \quad \text{———} (1{\cdot}36) \qquad \text{or} \quad U = \frac{P^2 L}{2AE}$$

But AL is the volume of the bar material and hence the above equation can be rearraged to give

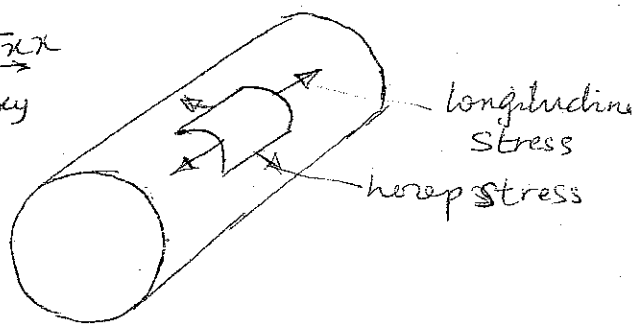
$$\frac{U}{AL} = \frac{\sigma^2}{2E} \quad \text{———} (1{\cdot}37)$$

Equation (1·27) can be stated; the strain energy per unit volume [usually called the resilience] in simple tension or compression is $\sigma^2/2E$.

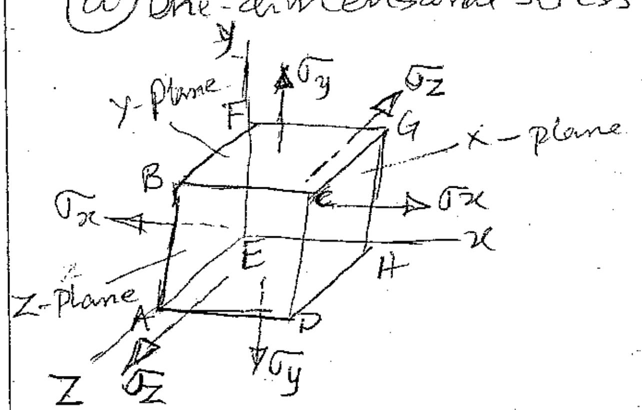
ANALYSIS OF PRINCIPAL PLANES, STRESSES AND STRAINS IN One-dimensional (Uni-axial), Two-dimensional (Uni-axial) and Three-dimensional (Multi-axial) stresses:



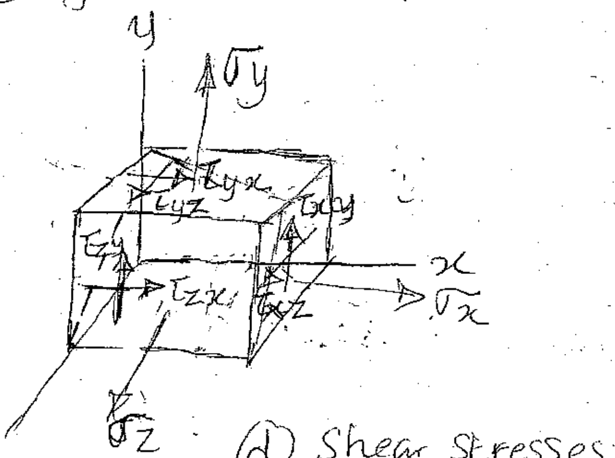
(a) One-dimensional stress



(b) Cylinder under pressure



(c) Normal or Direct stress



(d) Shear stresses

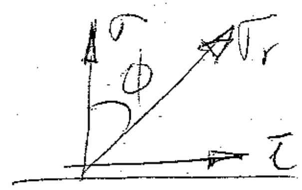
The above figures shows the representation of stresses in different dimensions. There must be uniformity in the naming of stresses assigned to them. The cubical element has six surfaces. Each surface is a plane and is named by the normal to its surface. For example in figure (c), the surface DCGH is named in the X-plane because X-axis is normal to it. By normal we mean the line emerging from the plane (outwards from the element). Note that ABFE is also an X-plane, but the negative direction of the X-axis is normal to it. This way the planes are named as follows

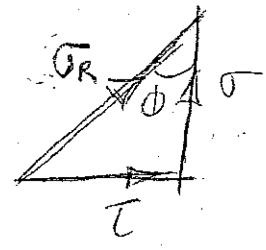
DCGH → + X-plane, BFGC → + Y-plane, ABCD → + Z-plane, ABFE → − X-plane, AEHD → − Y-plane, EFGH → − Z-plane.

Each surface can have one normal stress and two shear stresses (figure, d) σ_x or σ_{xx} shows the normal stress acting on the X-plane (1st subscript) and in the direction of the X-axis (2nd subscript). τ_{xy} means a shear stress on the X-plane (1st subscript) acting in the direction of the Y-axis (2nd subscript) and so on for the remaining normal and shear stresses in other axis/direction.

Stresses on oblique Section.

In many cases both direct and shear stress are brought into play, and the resultant stress across any section will be neither normal nor tangential to the plane. If σ_R is the resultant stress, making an angle ϕ with the normal to the plane on which it acts



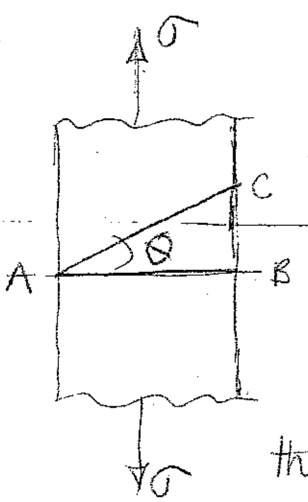


It is usual to calculate the normal (σ) and tangential (τ) components and by equilibrium.

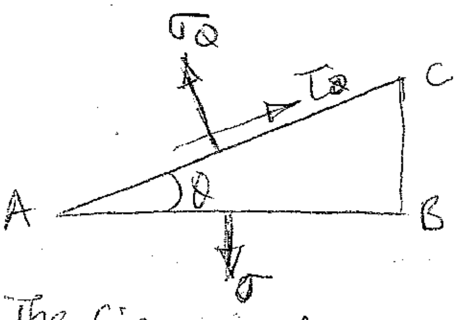
$$\sigma_R = \sqrt{\sigma^2 + \tau^2}$$

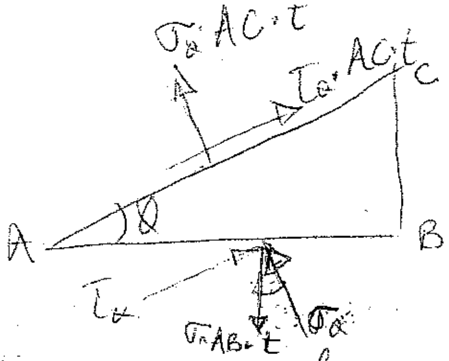
$$\phi = \tan^{-1}\left(\frac{\tau}{\sigma}\right)$$

Simple tension:



If a bar material is under the action of a tensile stress σ along its length, then any transverse section such as AB will be a pure normal stress acting on it. But the problem is to find the stress acting on any plane AC at an angle θ to AB. This stress will not be normal to the plane, and may be resolved into two components σ_θ and τ_θ.





The figure above shows the stresses acting on the three planes of the triangular prism ABC. There can be no stress on the plane BC, which is a longitudinal plane of the bar. Taking a thickness of the prism as t. The equation of equilibrium of forces are used to solve for σ_θ and τ_θ.

Resolving in the direction of σ_α:

$$\Sigma F_{\sigma_\alpha} = 0$$

$$\sigma_\alpha \cdot AC \cdot t - \sigma \cdot AB \cdot t \cos\alpha = 0$$

$$\sigma_\alpha \cdot AC \cdot t = \sigma \cdot AB \cdot t \cdot \cos\alpha$$

$$\sigma_\alpha = \sigma \cdot \frac{AB}{AC} \cdot \cos\alpha$$

$$\sigma_\alpha = \sigma \cos^2\alpha \longrightarrow (i)$$

Resolving in the direction of τ_α, $\Sigma F_{\tau_\alpha} = 0$

$$\tau_\alpha \cdot AC \cdot t - \sigma \cdot AB \cdot t \cdot \sin\alpha = 0$$

$$\tau_\alpha \cdot AC \cdot t = \sigma \cdot AB \cdot t \cdot \sin\alpha$$

$$\tau_\alpha = \sigma \cdot \frac{AB}{AC} \cdot \sin\alpha$$

$$= \sigma \cdot \cos\alpha \cdot \sin\alpha$$

$$\text{or } \tau_\alpha = \frac{1}{2}\sigma \sin 2\alpha \longrightarrow (ii)$$

$\sin 2\alpha = 2\cos\alpha\sin\alpha$

The resultant stress is given by

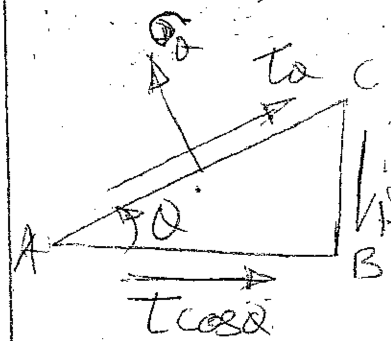
$$\sigma_r = \sqrt{(\sigma_\alpha^2 + \tau_\alpha^2)} = \sigma\sqrt{(\cos^4\alpha + \cos^2\alpha\sin^2\alpha)}$$

$$= \sigma\cos\alpha \longrightarrow (iii)$$

From these results it is seen that the maximum normal stress occurs at $\alpha = 0$ and this equal to the applied stress σ. The maximum shear stress occurs at $\alpha = 45^\circ$ and its magnitude is $\frac{1}{2}\sigma$. The variation of stress components with α is given by the above equations (i), (ii) and (iii). $\sigma_\alpha = 0$ when $\alpha = 90^\circ$ and $\tau_\alpha = 0$ when $\alpha = 0$. The resultant stress σ_r is maximum when $\alpha = 0$.

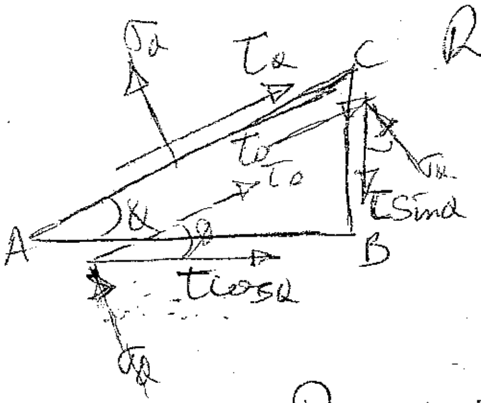
Another important result is that in simple tension or compression, the maximum shear stress is equal one-half of the applied stress and acts on planes at 45° to it.

Pure Shear Stress:



Let the stress on a given plane AB be a pure shear stress τ, then there is an equal complementary shear stress on the plane BC. The problem is to find the stress components σ_α and τ_α acting on the oblique plane AC at an angle α to AB.

It is clear from the previous results (simple tension - earlier analysed), that it was ~~independent of the thickness~~; of element considered and for the purpose of convenience we assume the hypotenuse to be of unit length, with this, the areas on which the stresses act are proportional to 1 (for AC), $\sin\alpha$ (for BC), and $\cos\alpha$ (for AB), and future figures will show the forces acting on such an element



Resolving forces in the direction of σ_α:

$$\sum F_{\sigma_\alpha} = 0$$

$$\sigma_\alpha - (\tau\cos\alpha)\sin\alpha - (\tau\sin\alpha)\cos\alpha = 0$$

$$\sigma_\alpha = (\tau\cos\alpha)\sin\alpha + (\tau\sin\alpha)\cos\alpha$$

$$= \tau\cdot\sin 2\alpha$$

Resolving forces in the direction of τ_α;

$$\sum F_{\tau_\alpha} = 0$$

$$\tau_\alpha - (\tau\sin\alpha)\sin\alpha + (\tau\cos\alpha)\cos\alpha = 0$$

$$\tau_\alpha = (\tau\sin\alpha)\sin\alpha - (\tau\cos\alpha)\cos\alpha$$

$$= \tau(\sin^2\alpha - \cos^2\alpha)$$

$$= -\tau\cos 2\alpha$$

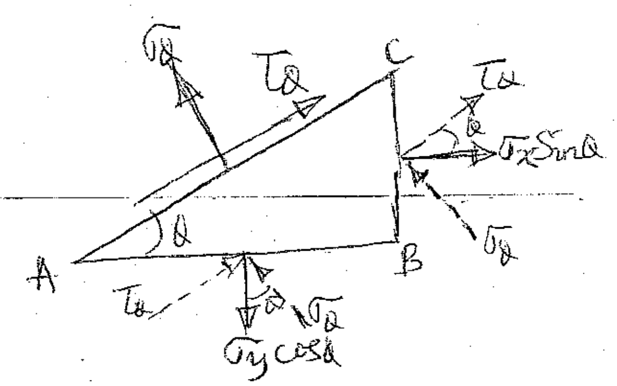
$$\sigma_r = \sqrt{(\sigma_\theta^2 + \tau_\theta^2)} = \sqrt{[(\tau \sin 2\theta)^2 + (-\tau \cos 2\theta)^2]}$$

$$= \tau, \text{ at } 2\theta$$

The result shows the normal component σ_θ has maximum and minimum values $+\tau$ (tension) and $-\tau$ (compression) on planes at $\pm 45°$ to the applied shear, and on these planes the tangential component τ_θ is zero.

This shows that at a point where there is pure shear stress on two given planes at right angles, the action across the planes of an element taken at $45°$ to the given planes is one of equal tension and compression.

Pure normal stresses on given planes.



Let the known stresses be σ_x on BC and σ_y on AB, then the forces on the element can be resolved in the direction of σ_θ and τ_θ

Resolving forces in the direction of σ_θ, we have

$$\sigma_\theta - (\sigma_y \cos\theta)\cos\theta - (\sigma_x \sin\theta)\sin\theta = 0$$

$$\sigma_\theta = \sigma_y \cos^2\theta + \sigma_x \sin^2\theta = \frac{1}{2}(\sigma_y + \sigma_x) + \frac{1}{2}(\sigma_y - \sigma_x)\cos 2\theta$$

Resolving forces in the direction of τ_θ,

$$\tau_\theta - (\sigma_y \cos\theta)\sin\theta + (\sigma_x \sin\theta)\cos\theta = 0$$

$$\tau_\theta = \sigma_y \cos\theta\sin\theta - \sigma_x \sin\theta\cos\theta$$

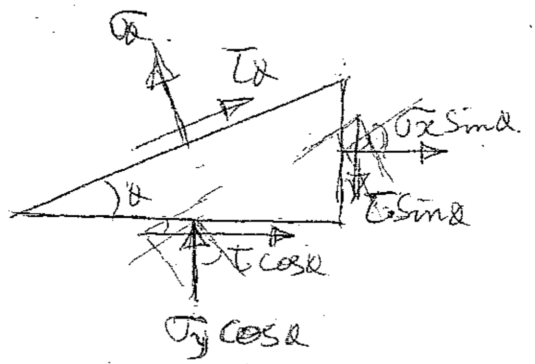
$$= \frac{1}{2}(\sigma_y - \sigma_x)\sin 2\theta$$

$\sin 2\theta = 2\cos\theta\sin\theta$

$\frac{1}{2}\sin 2\theta = \cos\theta\sin\theta$

General Two-dimensional Stress System

Let the stresses on the planes AB and BC be σ_y, σ_x and τ, as shown in the figure below



Resolving in the direction of σ_α:

$$\sigma_\alpha - (\sigma_y \cos\alpha)\cos\alpha - (\sigma_x \sin\alpha)\sin\alpha - (\tau\cos\alpha)\sin\alpha - (\tau \sin\alpha)\cos\alpha =$$

$$\sigma_\alpha = \sigma_y \cos^2\alpha + \sigma_x \sin^2\alpha + \tau\cos\alpha\sin\alpha + \tau\sin\alpha\cos\alpha$$

$$= \sigma_y\left(\frac{1+\cos 2\alpha}{2}\right) + \sigma_x\left(\frac{1-\cos 2\alpha}{2}\right) + \tau\sin 2\alpha$$

$$\sigma_\alpha = \frac{1}{2}(\sigma_y + \sigma_x) + \frac{1}{2}(\sigma_y - \sigma_x)\cos 2\alpha + \tau \sin 2\alpha \quad \text{———— (1)}$$

Resolving in the direction of τ_α;

$$\tau_\alpha - (\sigma_y \cos\alpha)\sin\alpha + (\sigma_x \sin\alpha)\cos\alpha - (\tau\sin\alpha)\sin\alpha + (\tau\cos\alpha)\cos\alpha = 0$$

$$\tau_\alpha = \sigma_y \cos\alpha\sin\alpha - \sigma_x \sin\alpha\cos\alpha + \tau\sin^2\alpha - \tau\cos^2\alpha$$

$$= \cos\alpha\sin\alpha(\sigma_y - \sigma_x) + \tau(\sin^2\alpha - \cos^2\alpha)$$

$$= \frac{1}{2}(\sigma_y - \sigma_x)\sin 2\alpha - \tau\cos 2\alpha \quad \text{———— (2)}$$

Principal Planes:

By defination the principal planes are those planes on which the shear stress is zero.
From equation (2) above in the general two-dimensional stress system.

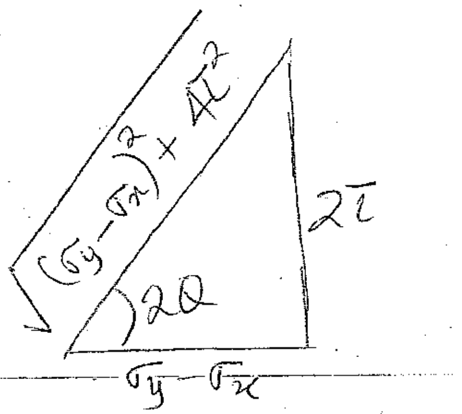
When $\tau_\theta = 0$, we have

$$\frac{1}{2}(\sigma_y - \sigma_x)\sin 2\theta - \tau\cos 2\theta = 0$$

This gives 2 values of 2θ, differing by 180° and hence two values of θ differing by 90° ie the principal planes are two planes at right-angles.

$$\frac{\sin 2\theta}{\cos 2\theta} = \frac{2\tau}{(\sigma_y - \sigma_x)}$$

$$\tan 2\theta = \frac{2\tau}{(\sigma_y - \sigma_x)}$$



$$\sin 2\theta = \pm \frac{2\tau}{\sqrt{[(\sigma_y - \sigma_x)^2 + 4\tau^2]}} \quad \text{—(3)}$$

$$\cos 2\theta = \pm \frac{(\sigma_y - \sigma_x)}{\sqrt{[(\sigma_y - \sigma_x)^2 + 4\tau^2]}} \quad \text{—(4)}$$

Principal stresses:

The stresses on the principal planes will be pure normal (tension or compression) and their value are called the principal stresses.
From equation (1), still under the general two-dimensional stress system.

Substituting for $\sin 2\theta$ (equation 3) and $\cos 2\theta$ (equation 4), we have

$$\text{Principal stresses} = \frac{1}{2}(\sigma_y + \sigma_x) \pm \frac{\frac{1}{2}(\sigma_y - \sigma_x)^2}{\sqrt{[(\sigma_y-\sigma_x)^2 + 4\tau^2]}} \pm \frac{\tau \cdot 2\tau}{\sqrt{[(\sigma_y-\sigma_x)^2 + 4\tau^2]}}$$

$$\sigma = \frac{1}{2}(\sigma_y + \sigma_x) \pm \frac{\frac{1}{2}[(\sigma_y-\sigma_x)^2 + 4\tau^2]}{\sqrt{[(\sigma_y-\sigma_x)^2 + 4\tau^2]}}$$

$$\sigma = \frac{1}{2}(\sigma_y + \sigma_x) \pm \frac{1}{2}\sqrt{[(\sigma_y-\sigma_x)^2 + 4\tau^2]} \quad \text{——— (5)}$$

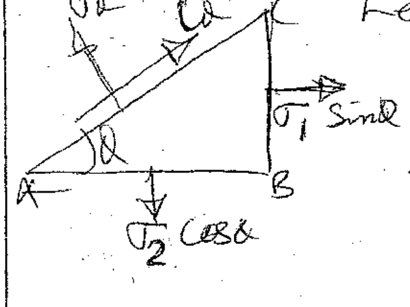
The importance of the principal stresses lies in the fact that they are the maximum and minimum value of normal stress in the two dimensions under consideration and when they are of opposite type they give the numerical values of the maximum tensile and compressive stresses. This can easily be verified by differentiating equation (1)

$$d\sigma_\theta / d\theta = -(\sigma_y - \sigma_x)\sin 2\theta + 2\tau \cos 2\theta$$

equating to zero for a maximum or minimum gives

$$\tan 2\theta = \frac{2\tau}{(\sigma_y - \sigma_x)}$$

Maximum Shear stress:



Let AB and BC be the principal planes and σ_1 and σ_2, the principal stresses, then

Resolving

$$\tau_\theta = (\sigma_2 \cos\theta)\sin\theta - (\sigma_1 \sin\theta)\cos\theta$$
$$= \frac{1}{2}(\sigma_2 - \sigma_1)\sin 2\theta$$

Hence the maximum shear stress occurs when $2\theta = 90$, ie on planes at $45°$ to the principal planes and its magnitude is

$$\tau_{max} = \frac{1}{2}(\sigma_2 - \sigma_1) \quad (6)$$

$$= \frac{1}{2}\sqrt{\left[(\sigma_x - \sigma_y)^2 + 4\tau^2\right]}$$ — Refer to Principal Stresses

This means that the maximum shear stress is one-half the algebraic difference between the principal stresses. The same result could be obtained by differentiating equation 2 - under the general two-dimensional stress system.

It is important to mention that, for all solids being of three dimensions, there must be three (3) principal stresses, although in many cases the third principal stress is zero. In calculating the maximum shear stress by taking one-half the algebraic difference between the principal stresses, the zero principal stresses ~~will be of importance~~ if the other two are of the same type.

MOHR'S CIRCLE.

Based on the two equation σ_θ and τ_θ [equations 1 and 2, derived earlier in the general two-dimensional stress system], Otto Mohr a German engineer, derived a very elegant graphical procedure for the analysis of plane stress. The diagram obtained gives a vivid, visual picture of the interpretation of the equations.

Deriving the Equation for Mohr's Circle

$$\sigma_\theta = \frac{1}{2}(\sigma_y+\sigma_x) + \frac{1}{2}(\sigma_y-\sigma_x)\cos 2\theta + \tau\sin 2\theta$$

and $$\tau_\theta = \frac{1}{2}(\sigma_y-\sigma_x)\sin 2\theta - \tau\cos 2\theta$$

It must be noted that, in the above equations, σ_x, σ_y and τ are known quantities, and σ_θ and τ_θ are the normal and shear stresses on any arbitrary plane identified by θ.

Rearranging and squaring these equations, we get,

$$\sigma_\theta - \frac{\sigma_y+\sigma_x}{2} = \left(\frac{\sigma_y-\sigma_x}{2}\right)\cos 2\theta + \tau\sin 2\theta$$

$$\left[\sigma_\theta - \frac{\sigma_y+\sigma_x}{2}\right]^2 = \left[\left(\frac{\sigma_y-\sigma_x}{2}\right)\cos 2\theta + \tau\sin 2\theta\right]^2$$

$$\left[\sigma_\theta - \frac{\sigma_y+\sigma_x}{2}\right]^2 = \left(\frac{\sigma_y-\sigma_x}{2}\right)^2(\cos 2\theta)^2 + 2\left(\frac{\sigma_y-\sigma_x}{2}\right)\tau\cos 2\theta\sin 2\theta + \tau^2(\sin 2\theta)^2 \quad \text{— (8)}$$

Similarly

$$\tau_\theta^2 = \left[\left(\frac{\sigma_y-\sigma_x}{2}\right)\sin 2\theta - \tau\cos 2\theta\right]^2$$

$$\tau_\theta^2 = \left(\frac{\sigma_y-\sigma_x}{2}\right)^2(\sin 2\theta)^2 - 2\left(\frac{\sigma_y-\sigma_x}{2}\right)\tau\sin 2\theta\cos 2\theta + \tau^2(\cos 2\theta)^2$$

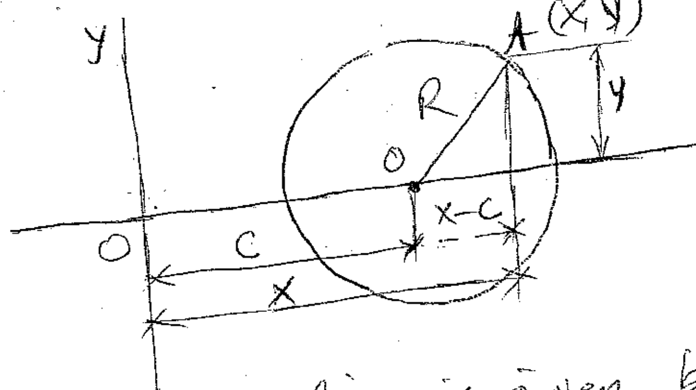
/27

Adding equations (8) and (9), we get

$$\left[\sigma_\theta - \frac{\sigma_y + \sigma_x}{2}\right]^2 + \tau_\theta^2 = \left(\frac{\sigma_y - \sigma_x}{2}\right)^2 + \tau^2 \qquad (10)$$

Equation is known as the <u>equation for Mohr's circle.</u>

Considering the equation for a standard circle as shown in the figure below: we see that (X, Y) are the coordinates of a point A, on a circle whose radius is R and whose centre is shifted by C from the origin O along the X-axis

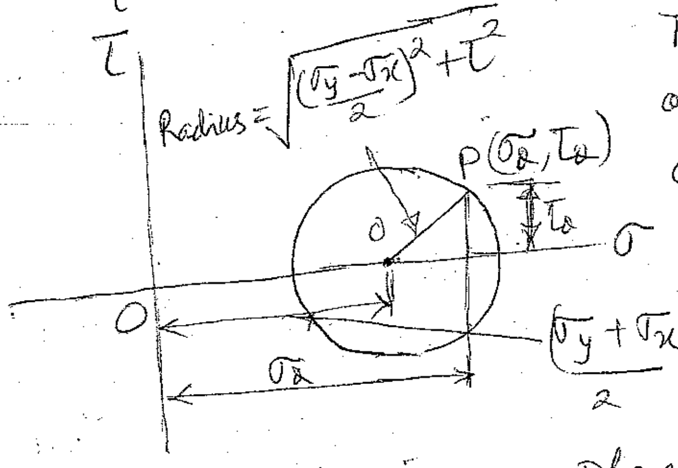


The standard equation is given by;

$$(X-C)^2 + Y^2 = R^2 \qquad (11)$$

The equation for Mohr's circle is identical to the equation for the standard circle.

The figure is the equation of a circle whose points have coordinates $(\sigma_\theta, \tau_\theta)$ at P, and whose radius is indicated and its centre is shifted along the σ_θ axis as indicated also.

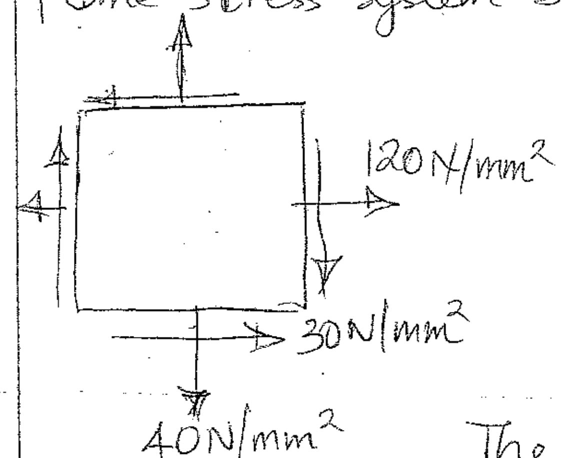


This means that given a plane stress system, the stresses of any inclined plane σ_θ and τ_θ can be represented as the coordinates of a point on a circle which is

Set as σ, and τ axes system.

Drawing Mohr's circle:

One of the method to draw Mohr's circle is to calculate the coordinates of the centre and radius; thus for the plane stress system shown below.

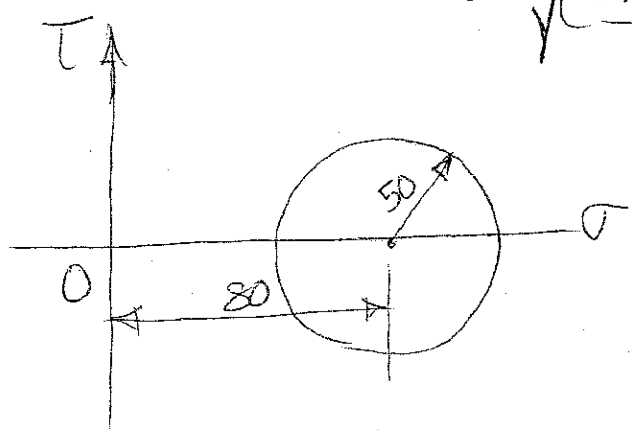


$$\frac{\sigma_x + \sigma_y}{2} = \frac{120 + 40}{2} = 80 \text{ units}$$

80 units is the shift centre.
The coordinates of the centre of Mohr's circle are $(80, 0)$

The radius of the circle is given by $\sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \sqrt{\left(\frac{120-40}{2}\right)^2 + 30^2}$

$= 50$ units

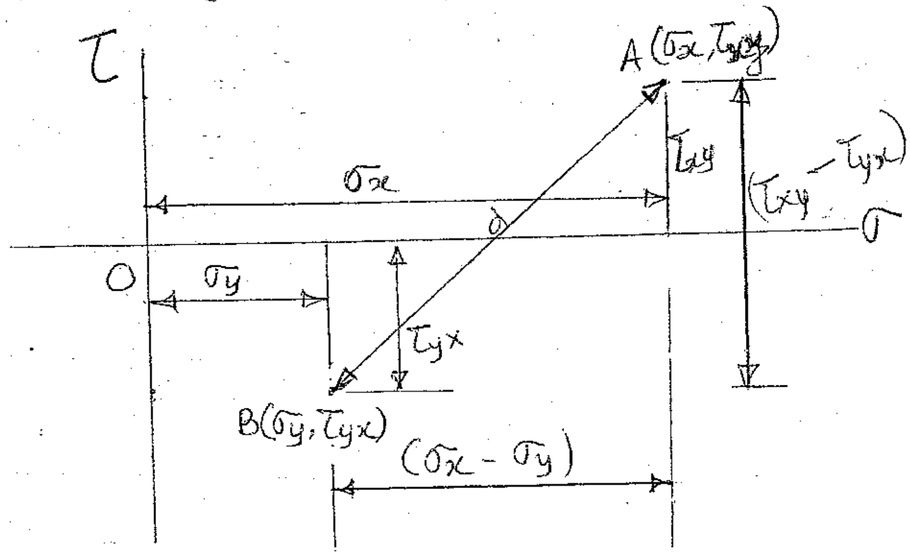


The Mohr's circle is drawn as indicated

It should be noted that the centre of Mohr's circle lies on the σ-axis.
Any point on the circle has coordinates $(\sigma_\alpha, \tau_\alpha)$ where σ_α is the normal stress and τ_α is the shear stress on a plane by the inclination of the normal to the X-axis, α.
Another method ~~for~~ drawing Mohr's circle is based on the fact that stresses on orthogonal planes plot as diametrically opposite points on the circle. To prove this, note that, the distance between two points represented by these two points must be equal to the diameter of the circle.

From the figure below, the distance between points represented by (σ_x, τ_{xy}) and (σ_y, τ_{xy}) may be worked out from:

$$d^2 = (\sigma_x - \sigma_y)^2 + (\tau_{xy} - \tau_{yx})^2$$



The radius of Mohr's circle as proved earlier may be obtained from

$$R^2 = \left(\frac{\sigma_x - \sigma_y}{2}\right) + \tau_{xy}^2$$

Since $\tau_{yx} = - \tau_{xy}$

If D is the diameter of the Mohr's circle, then

$$\left(\frac{D}{2}\right)^2 = \left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2$$

$$D^2 = (\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2$$

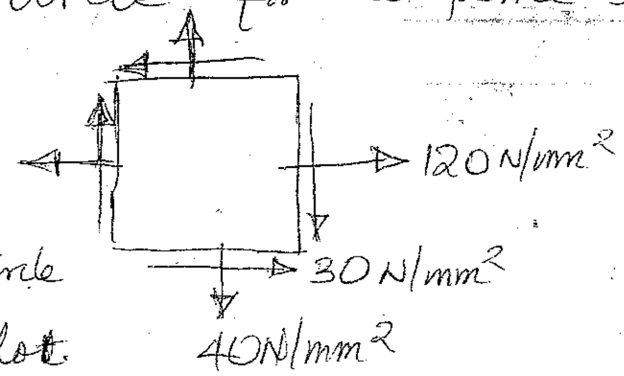
Noting that $\tau_{yx} = \tau_{xy}$, we have

$$d^2 = (\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2$$

which is the same as the square of the diameter of Mohr's circle.

Example: 1.

Draw Mohr's circle for the plane stress system shown below:



To draw Mohr's circle

i) Select a scale to plot the stresses

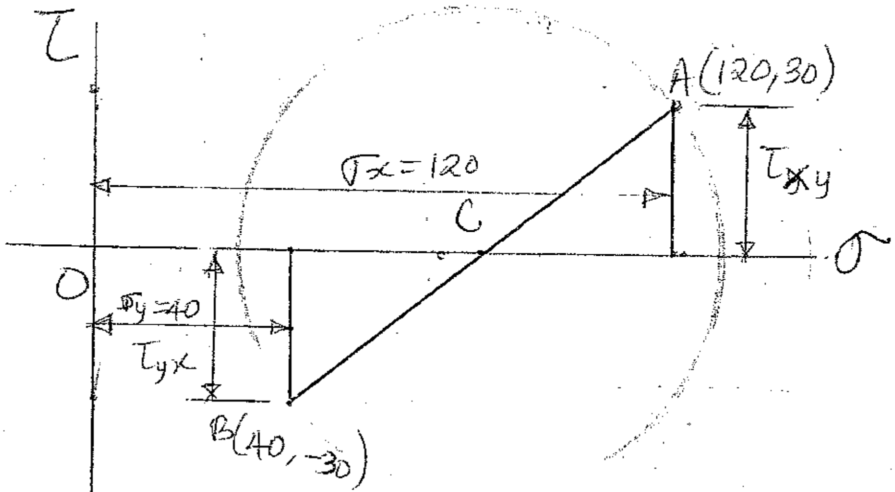
ii) Draw the σ-τ axes conveniently

iii) plot two point with coordinates A(120, 30) and B(40, -30)

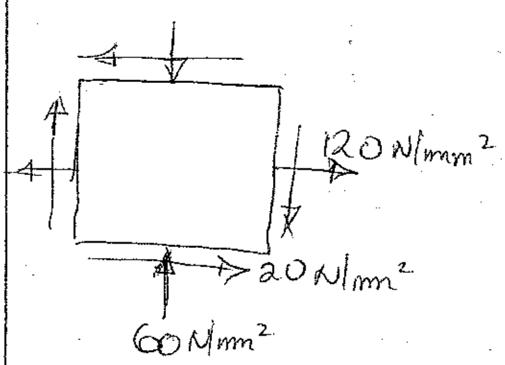
iv) Join the two points - where the line cuts the σ-axis is the centre of Mohr's circle

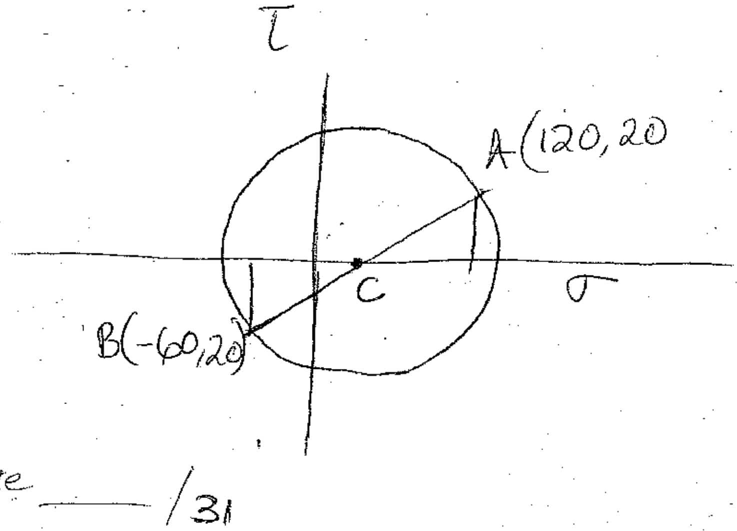
v) with C as centre and CA or CB as radius, draw a circle which is Mohr's circle

Scale: 1cm = 15N/



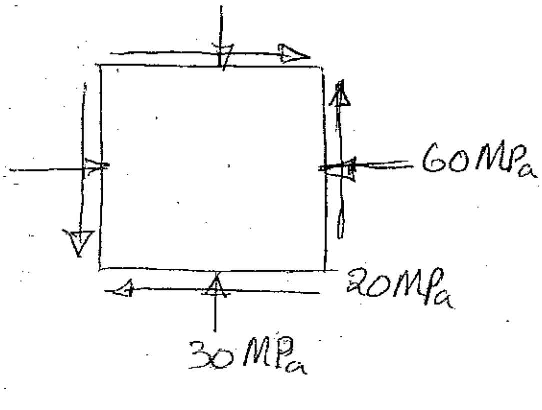
Example 2.

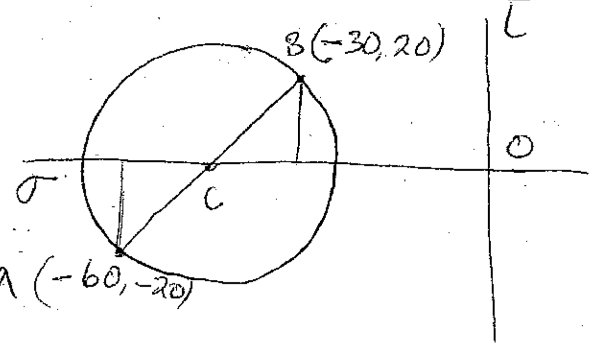




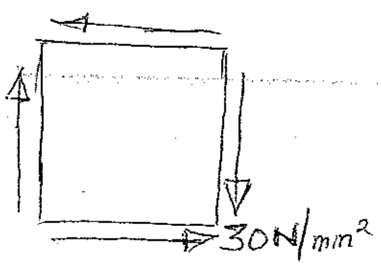
Note: σ_y is compressive

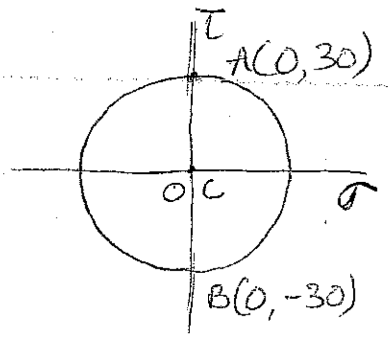
Example 3: A Case of compressive stress should be plotted to the left of the origin





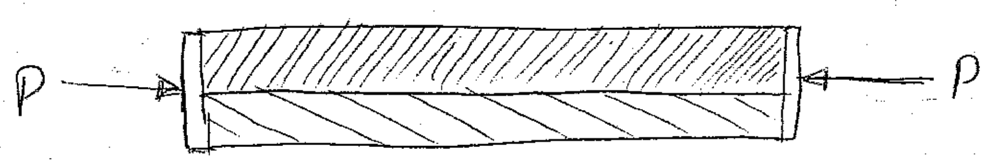
Example 4: A Case of pure Shear, the centre of Mohr's Circle coincides with the origin.





COMPOSITE BARS OR SECTIONS.

A composite bars/Section consists of two or more bars in parallel, usually of different materials bonded together rigidly. Any tensile or compressive acting on the bar, the resulting deformation is shared by the materials forming the Section. The figure below shows a composite bar is made up of a rod of Area A_1 and Modulus E_1 and another rod of equal length of area A_2 and modulus E_2. If a compressive load P is applied to the composite bar.



Since the two rods are of the same initial length & must remain together, then the strain in each part must be the same. The total load carried is P and let the shared load be P_1 and P_2.

Since strain, $\varepsilon = \frac{P}{AE}$

We can state that; $\frac{P_1}{A_1E_1} = \frac{P_2}{A_2E_2}$ —— (i) as the compatibility

Condition and for equilibrium $P = P_1 + P_2$ —— (ii)

from equation $P_2 = \frac{A_2E_2}{A_1E_1} \cdot P_1$ —— (iii)

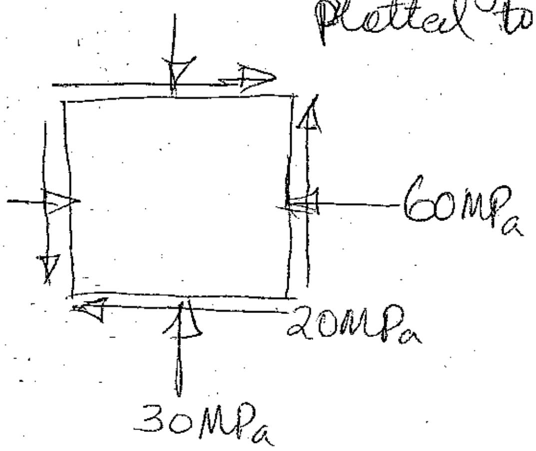
Substituting equation (iii) in (ii), we have

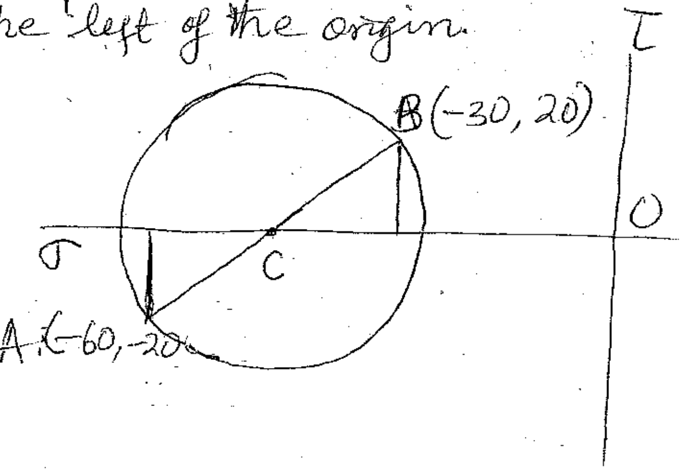
$$P = P_1\left(1 + \frac{A_2E_2}{A_1E_1}\right)$$

$$\text{or} \quad P_1 = \frac{P \cdot A_1E_1}{A_1E_1 + A_2E_2}$$

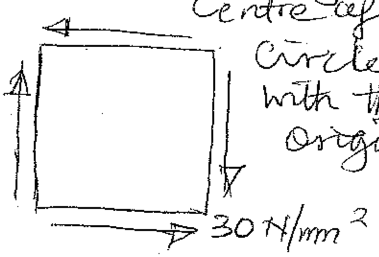
$$\text{illy} \quad P_2 = \frac{P \cdot A_2E_2}{A_1E_1 + A_2E_2}$$

Example 3 : A case of compressive stress should be plotted to the left of the origin.

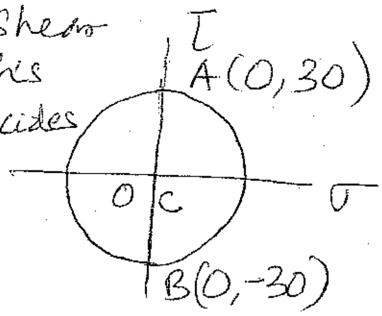




Example 4 : A case of pure shear

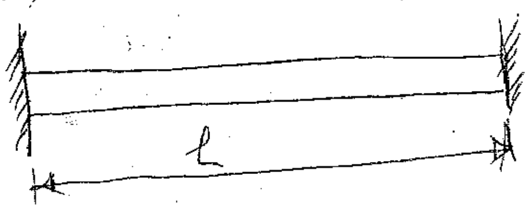


Centre of Mohr's circle coincides with the origin.



STRESSES DUE TO TEMPERATURE CHANGE.

When a body is subjected to temperature change, it elongates if temperature rises and shortens if the temperature decreases. Stresses are caused in a material if the natural expansion or contraction is prevented. The process of expansion on heating and contraction on cooling is known as thermal expansion or contraction. If the material is constrained and is not fully or partly free to change its length, the stresses are developed in the material, called temperature or thermal stresses.



$$\varepsilon_x = \frac{\sigma_x}{E} - \nu\frac{\sigma_y}{E} - \nu\frac{\sigma_z}{E}$$

$$\varepsilon_y = \frac{\sigma_y}{E} - \frac{\sigma_z}{E} - \nu\frac{\sigma_x}{E}$$

$$\varepsilon_z = \frac{\sigma_z}{E} - \frac{\sigma_x}{E} - \frac{\sigma_y}{E}$$

$$\varepsilon_v = \frac{\sigma_x}{E} - \nu\frac{\sigma_y}{E} - \nu\frac{\sigma_z}{E} + \frac{\sigma_y}{E} - \nu\frac{\sigma_z}{E} - \nu\frac{\sigma_x}{E} - \nu\frac{\sigma_z}{E} - \nu\frac{\sigma_x}{E} - \nu\frac{\sigma_y}{E}$$